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Academic Year: 2025-26 (Even Semester)

Major Project Phase-I (CSEP23605)

Review 0 - Synopsis Submission

Date	03/03/2026	**Domain**	Cyber Security
Group Number	G40	**Project ID**	P52
Project Title	AI- Powered Cybersecurity Platform for Real-Time Intrusion Detection		

Problem Description:

AI-Driven Cybersecurity Platform : Cyber attacks are growing in scale and complexity, while traditional security tools struggle to detect evolving threats. Manual monitoring is impractical due to massive data volumes. There is a need for intelligent, automated systems that can learn from attack patterns and provide proactive defense.

Objectives:

- Develop an AI-based model for anomaly and intrusion detection.
- Analyze malware and phishing patterns using machine learning.
- Build a scalable framework for real-time cyber threat prediction.
- Automate Threat Response and Alert Generation

Literature survey Completed So far and Gap Identified through the Literature Survey

Sl. #	Title of the paper and year	Methodology	Advantages	Disadvantages
1	Cantone, Marrocco & Bria (2024) -- Machine Learning in Network Intrusion Detection: A Cross-Dataset Generalization Study	Evaluates ML-based IDS models across multiple benchmark datasets to measure cross-dataset generalization (base paper for this project's own IEEE submission)	Directly documents the accuracy collapse when models are tested outside their training distribution	Does not propose an adaptive retraining fix for the gap it identifies
2	Zouhri, Idri & Ratnani (2024) -- Evaluating the Impact of Filter-Based Feature Selection in IDS	Compares filter-based feature selection techniques (e.g. ANOVA, mutual information) for IDS	Reduces feature dimensionality while preserving detection accuracy	Feature selection tuned per-dataset; generalization not addressed

3	Gupta et al. (2025) -- Generative Active Adaptation for Drifting and Imbalanced Network Intrusion Detection	Generative + active-learning adaptation to handle traffic drift and class imbalance	Directly targets model drift under evolving, imbalanced traffic	Adds generative-model training overhead and complexity
4	Elangovan et al. (2026) -- Cross-Dataset Temporal and Semantic Generalization of IDS Models for the Future Internet	Studies temporal and semantic drift across IDS datasets over time	Explicit framework for temporal generalization	Evaluated on benchmark datasets only, not live captured traffic
5	Sharafaldin, Lashkari & Ghorbani (2018) -- Toward Generating a New Intrusion Detection Dataset and Traffic Characterization	CICFlowMeter-based flow generation and labeled attack-scenario capture (the CIC-IDS-2017 dataset this project trains on)	Large-scale, realistic, publicly available labeled benchmark	Static snapshot -- doesn't capture the live-traffic drift this project specifically targets
6	Chen & Guestrin (2016) -- XGBoost: A Scalable Tree Boosting System	Gradient-boosted decision tree algorithm (the core classifier this project uses)	High accuracy, fast, handles imbalanced tabular flow data well	Not IDS-specific; needs feature engineering and tuning for network traffic
7	Viegas, Santin & Oliveira (2017) -- Toward a Reliable Anomaly-Based Intrusion Detection in Real-World Environments	Anomaly-based IDS evaluated under real-world deployment constraints	Early direct study of the benchmark-vs-real-world reliability gap	Predates modern gradient-boosted and explainable approaches
8	Layeghy & Portmann (2023) -- Explainable Cross-Domain Evaluation of ML-Based Network Intrusion Detection Systems	Applies SHAP-style explainability analysis to cross-domain IDS evaluation	Combines generalization study with interpretability	Explainability used for offline analysis only, not wired into a live response pipeline
9	Thanh (2026) -- A Transfer-Aware, Deployment-Oriented Evaluation Framework for NetFlow-Based IDS (TAN-IDS)	Deployment-oriented, transfer-aware evaluation framework for NetFlow-based IDS	Explicitly deployment-oriented rather than benchmark-only	Framework-level contribution; no adaptive retraining loop
10	Lundberg & Lee (2017) -- A Unified Approach to Interpreting Model Predictions (SHAP)	Shapley-value-based unified feature attribution framework, used by this project for explainability	Model-agnostic, theoretically grounded feature importance	Computationally expensive at scale; not IDS-specific

Panel Members' Remarks (if any):	
Status:	○ **Approved** ○ **Rejected** ○ **Revision needed**

	Name	**Signature**
Panel Member 1	Prof.B S Umashankar	
Panel Member 2	Prof.Ravi Babu G	
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